

Anthony Chen

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Education

University of California, San Diego | GPA: 3.98

Expected Graduation: Jun 2028

Majors: Bioinformatics and Biochemistry

Minor: Computer Science

- Relevant Courses: Object-Oriented Programming in Java, Basic Data Structures and Object-Oriented Design, Linear Algebra, Discrete Mathematics, Multivariate Calculus and Analytic Geometry, Mathematics for Algorithms and Systems, Probability and Statistics for Bioinformatics, Data Analysis and Design for Biologists

Work Experience

Dr. Ruben Abagyan Lab

San Diego, CA

Undergraduate Researcher

Dec 2025 - Present

- Bash scripting with the Protein Data Bank to extract biologically relevant information from specific target proteins for multi-target pharmacology profiling

Activities

Undergraduate Bioinformatics Club

Member

- Developed a **Cerebrospinal Fluid Disease ML Classifier** with my team in the 2025 bioinformatics hackathon by web scraping **1000+ single-cell RNA sequencing cerebrospinal fluid data from Gene Expression Omnibus datasets**, merging this data into a high-dimensional matrix using the **Anndata library**, visualizing this dataset by using the **UMAP** technique with the **Scanpy library**, and feeding the data into a **scikit-learn random forest classifier model** to predict the disease of a patient with **~70% accuracy**
- Utilized **Principal Component Analysis (PCA)** to reduce dimensionality of high-dimensional data and visualize differences between samples based on clusters

Technical Skills

Computational: Java, Python, Linux, R, MATLAB, Excel, MS Word

Leadership Experience

Forte Music Club

San Francisco, CA

Founder

Jan 2023 – Aug 2024

- Contacted various senior centers and school musicians to coordinate **5 performances**
- Created **Google forms and a Discord server** to coordinate optimal performance/practice dates
- Led and played in a jazz ensemble that performed at senior centers

Marching Band

San Francisco, CA

Section Leader

Oct 2023 – Dec 24

- Led and demonstrated marching techniques to the trumpet section **3 days** a week
- Held music and social sectionals outside of practices to teach the show music and promote bonding within the trumpet section
- Organized and collaborated with other section leaders to help competitions run smoothly